Joonsuk Bae

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Lead analyzer of the first ALICE Run 3 charged-particle jet measurement. Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.

Education

Ph.D. Candidate, Physics — Sungkyunkwan University 2022.03 – expected 2026.08

Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp collisions at* $\sqrt{s} = 13.6$ *TeV with ALICE.*

B.S., Physics — Sungkyunkwan University 2021.02

Appointments and Collaboration Membership

ALICE Collaboration, CERN, Geneva	2022.10 – present
ePIC Collaboration, Brookhaven National Laboratory	2024.02 – present
LAMPS Collaboration, RAON, Korea	2022.03 – 2024.04

Awards and Fellowships

- **BK21 FOUR Graduate Scholarship** — Sungkyunkwan University Physics BK21 sponsored unit (Korean Ministry of Education), 2022 – present.
- **Conference travel support** — KoALICE / NRF Korea–CERN / SKKU graduate school support across 2023–2025 international conferences.

Research Experience

Lead analyzer, ALICE Run 3 charged-particle jets (PWG-JE) — CERN 2023.04 – present

- Leading the first ALICE Run 3 charged-particle jet cross-section measurement at $\sqrt{s} = 13.6$ TeV end-to-end — O^{2}Physics task development, QA, unfolding, full systematic-uncertainty program, and preliminary release. PWG-JE-approved results establish the Run 3 inclusive charged-jet baseline.
- **Tracking-efficiency systematics** (PWG-JE Run 3 tracking-systematics task force, convened 2026.02 by H. Caines and L. Havener, Yale; biweekly meetings) — pp $\sqrt{s} = 13.6$ TeV lead analyzer. Derived the total charged-particle tracking-efficiency systematic for charged-jet analyses ($\sim$1.5% across the dominant p_{T} range; p_{T}-binned), including extension to the 2022 and 2024 periods. Contributing p_{T}-, η-, and ϕ-dependent ITS-TPC matching studies; ITS-TPC matching analysis code adopted by J. Lee for PbPb tracking studies. Findings feed the Hard Probes 2026 systematic-uncertainty proposal. Prior to the task force, PWG-JE jet analyses inherited the DPG-Tracking Run 1/2 central value without dedicated Run 3 validation.
- **Trigger-level inefficiency in the new readout** — quantified bunch-crossing and track-matching inefficiencies introduced by the continuous-readout TPC and asynchronous reconstruction (each with its own time-frame); derived signal-loss corrections now applied across PWG-JE charged-jet analyses.
- **Track p_{T} resolution** — studied data–MC resolution mismatch via covariance-matrix propagation, MC residuals, and V0 mass-resolution methods. Tuned the central TrackTuner with a p_{T}-dependent smearing now standard for Run 3 charged-jet systematics, and patched the central reconstruction paths where Run 1/2 conventions had been silently propagated.
- **Cross-section normalization** — established the Run 3 charged-jet normalization based on the TVX visible cross section under continuous readout; sole author of `jetCrossSectionEfficiency.cxx` (PR #14057, merged) used by all PWG-JE charged-jet analyses for normalization. Provided the normalization factor and QA for the ALICE Run 3 O–O charged-jet measurement led by a junior co-analyzer (W. Ham) in the SKKU group.
- Presented preliminary results at **EPS-HEP 2025** (on behalf of the ALICE Collaboration; PWG-JE merge talk), **Quark Matter 2025**, and **Hard Probes 2024**; regular analyzer in PWG-JE working-group meetings (50+

internal presentations since 2022).

- Extending the program to semi-inclusive hadron+jet measurements, R-dependent jet suppression, and jet substructure in small systems (high-multiplicity pp, O–O, Ne–Ne) as the basis of the thesis.

Detector physicist, ePIC Barrel Imaging Calorimeter (BIC) — Korea-BIC consortium 2024.02 – present

- Led PMT-module QA prior to test beams — cosmic-ray and source signal characterization, timing-coincidence checks, optical-coupling stability, and per-module acceptance criteria.
- Lead offline analyst across two test-beam campaigns of the Pb/scintillating-fiber prototype: **KEK AF–AR** (2025.03) and **CERN PS T10** (2025.07); produced the first physics output of the Korea-BIC prototype (energy response, resolution, linearity).
- Lead JANA2 / EIC-recon reconstruction studies determining energy- and η-dependent sampling fractions; contributed Geant4 prototype modeling.
- Module assembly and integration — Pb/scintillating-fiber bundle alignment, polishing, PMT/SiPM coupling.

Detector R&D, LAMPS Forward Tracker — RAON, Korea 2022.03 – 2024.04

- Early-stage design and Geant4 simulation of an MPPC–scintillator-plane forward tracking detector; sensor surveys, specifications, and tracking-concept feasibility studies.

Selected Publications

Full publication list: inspirehep.net/authors/2117232. ALICE Collaboration co-author since 2022.10 (**109 collaboration papers**).

ALICE results led by the author (in preparation / internal review)

- FLAGSHIP ALICE Collaboration, *Inclusive and semi-inclusive charged-particle jet cross sections in pp collisions at* $\sqrt{s} = 13.6$ *TeV* (in preparation, target 2026). *First ALICE Run 3 charged-jet publication.* *[lead analyzer; full responsibility for analysis, systematics, unfolding, and paper draft]*
- ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at* $\sqrt{s} = 13.6$ *TeV* — thesis analysis, foreseen 2026. *[lead analyzer; sole analyst of the R-dependence component]*

ePIC / EIC detector R&D and design (peer-reviewed)

- J. K. Adkins *et al.* (ECCE Collaboration), “Design of the ECCE detector for the Electron-Ion Collider,” *Nucl. Instrum. Meth. A* **1073** (2025) 170240. > 280 *citations.* *[co-author; ECCE consortium member]*
- S. Y. Jang, G. An, **J. Bae** *et al.*, “Performance of dual-readout calorimeters for various absorbers,” *Nucl. Instrum. Meth. A* (2026), DOI `10.1016/j.nima.2026.171598`. *[co-author; dual-readout R&D for e^+e^- colliders]*
- G. Cho, S. Kim, **J. Bae** *et al.*, “Beam tests of the copper-based dual-readout calorimeter to measure electromagnetic performance for future e^+e^- colliders,” *J. Subatom. Part. Cosmol.* **3** (2025) 100021. *[co-author; module construction and beam-test data taking]*
- ECCE Consortium, “AI-assisted optimization of the ECCE tracking system at the Electron-Ion Collider,” *Nucl. Instrum. Meth. A* (2023), DOI `10.1016/j.nima.2022.167748`. *[co-author; ECCE consortium member]*
- ECCE Consortium, “Open heavy-flavor studies for the ECCE detector at the Electron-Ion Collider,” `arXiv:2207.10632` (2022). *[co-author; preprint]*

LAMPS detector R&D (peer-reviewed)

- Y. J. Kim, D. S. Ahn, J. K. Ahn *et al.* (LAMPS), “Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON,” *J. Korean Phys. Soc.* **87** (2025) 670–682. *[co-author; Time-of-Flight detector maintenance and on-site checks]*

- Y. Bae *et al.* (LAMPS), “Performance test of prototype LAMPS Start Counter using proton beams at 100 MeV,” *Nucl. Instrum. Meth. A* (2025), DOI `10.1016/j.nima.2025.170827`. *[co-author; Start-Counter prototype testing]*
- Y. Cheon *et al.* (LAMPS), “Characteristics of the prototype LAMPS TPC for low-energy nuclear collisions at RAON,” *Nucl. Instrum. Meth. A* **1066** (2024) 169610. *[co-author; TPC prototype characterization]*
- H. Kim *et al.* (LAMPS), “Performance of the prototype beam drift chamber for LAMPS at RAON with proton and ^{12}C beams,” *JINST* **19** (2024) P12008; `arXiv:2412.08662`. *[co-author; forward-tracking R&D and prototype performance studies]*
- LAMPS Collaboration, “Status of large acceptance multi-purpose spectrometer (LAMPS) at RAON,” *AIP Conf. Proc.* **2319** (2021) 080002. *[co-author; SiPM and scintillator-based detector R&D]*

Other peer-reviewed

- “Simulation study for the energy and position reconstruction performances of the beam monitoring system of Carbon Ion Radiation Therapy using Geant4,” *PLoS ONE* (2025), DOI `10.1371/journal.pone.0313862`. *[co-author; Geant4 simulation contributions]*

Submitted (detector)

- Y. Hong, J. Bok, G. An, **J. Bae** *et al.*, “Test-beam performance of the AstroPix silicon sensor for imaging calorimetry,” submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00572). Corresponding: Prof. Sanghoon Lim. *[co-author; ePIC BIC silicon-layer test-beam analysis]*
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae** *et al.*, “Beam test of a Pb/SciFi prototype for the Barrel Imaging Calorimeter at the Electron-Ion Collider,” submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00482; also submitted to JKPS). *[co-author; Pb/SciFi prototype beam-test data]*

In preparation (detector)

- Korea-BIC Collaboration, *Beam-test characterization of a Pb/scintillating-fiber imaging calorimeter prototype for the EIC* (three papers in preparation, 2026). *[primary data analyst across all three; co-lead author on the energy-response paper]*

Software and Computing

ALICE O^{2}Physics — AliceO2Group/O2Physics

- **Central framework patch in** `Common/` (PR #15174, merged) — fixed stale Q/p_{T} values in the graph-based TrackTuner momentum smearing; affects tracking systematics for all downstream analyses using TrackTuner.
- **Sole author of** `jetCrossSectionEfficiency.cxx` (PR #14057, merged) — the PWG-JE task used by all Run 3 charged-jet analyses to compute the cross-section normalization at particle level.
- 35 pull requests / 19 merged (2023 – present) covering jet finding, response matrices, systematic-uncertainty pipelines, hadronic-rate fixes, and central reconstruction; code review on 5 community PRs across PWG-JE and Common.

Invited and Contributed Talks

Summary since 2022: 3 invited workshops *[external]* · 5 invited collaboration-meeting talks *[internal]* · 6 international conference contributions (one on behalf of the ALICE Collaboration at EPS-HEP 2025) · 7 domestic (KPS) · PWG-JE internal: 50+ .

Invited Talks (external workshops, seminars)

- **France–Korea Particle Physics Network (FKPPN) Workshop**, Inha University — *Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3* — 2025.12
- **1st Heavy-Ion Meeting (HIM 2025)**, APCTP — *From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies* — 2025.05
- **FKPPN Workshop**, Inha University — *Charged-jet studies in ALICE Run 3* — 2023.11

Invited Collaboration-Meeting Talks (internal)

- **Korea-BIC 2025 Summer Workshop** — *Data analysis of the recent beam test at CERN* — 2025.09
- **KoALICE National Workshop 2024** — *Jet analysis status: first look at the charged-particle jet spectrum in Run 3 pp collisions* — 2025.01
- **ALICE Physics Week 2024** — *Charged-particle jet production in Run 3* — 2024.12
- **KoALICE National Workshop 2023** — *Charged-jet O^2 MC and data QA* — 2024.01
- **KoALICE National Workshop 2022** — *Dijet studies with ALICE at the LHC* — 2023.01

Contributed Talks and Posters

- **EPS-HEP 2025**, Marseille — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE [on behalf of the ALICE Collaboration; PWG-JE merge talk]* — 2025.07
- **INPC 2025**, Daejeon — *Charged-particle jet and jet quenching measurement in pp collisions at $\sqrt{s} = 13.6$ TeV during LHC Run 3 with ALICE [international, talk]* — 2025.05
- **Quark Matter 2025**, Frankfurt — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE [international, poster]* — 2025.04
- **Hard Probes 2024**, Nagasaki — *First measurements of charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE [international, poster]* — 2024.09
- **Quark Matter 2023**, Houston — *Charged-particle multiplicity distribution in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE [international, poster]* — 2023.09
- **ATHIC 2023** (9th Asian Triangle Heavy-Ion Conf.), Incheon — *Dijet studies at the LHC [international, poster]* — 2023.04
- **Korean Physical Society meetings** (Spring/Fall 2022 – 2025) *[domestic]* — 5 contributed talks and 2 posters on dijet studies (ALICE), charged-particle jet production in pp at $\sqrt{s} = 13.6$ TeV (Run 3), and ePIC BIC prototype QC.

Test-beam and Detector Activities

ALICE ITS2 alignment monitoring at CTF level (Run 3)

- Analyzed compressed-time-frame (CTF) data — below the standard AO2D analyzer interface — to provide period-by-period tracking-performance diagnostics.

ePIC Barrel Imaging Calorimeter test beams

- **CERN PS T10** (2025.07) — on-site analysis lead; energy response and resolution.
- **KEK AF–AR** (2025.03) — DAQ, equalization, prompt analysis.
- **Korea-BIC module production** (2024 – 2025) — Pb/scintillating-fiber bundle assembly, polishing, and PMT/SiPM integration; PMT-level QA across all prototype modules.

Service and Professional Activities

- **PWG-JE Run 3 tracking-systematics task force** (convened 2026.02 by H. Caines and L. Havener, Yale; biweekly) — pp 13.6 TeV lead analyzer; ITS-TPC matching code adopted for PbPb tracking by other group members (see Research Experience).
- **Co-author, ALICE Internal Analysis Note** (PWG-JE) — ALICE Run 3 O–O charged-jet measurement; contribution: cross-section normalization factor derivation and QA.
- **ALICE ITS2 alignment monitoring** — service work at CTF level for tracking-performance diagnostics (see Test-beam and Detector Activities).
- **O^{2}Physics community** — code review on 5 community pull requests across PWG-JE and Common.
- **ePIC Korea-BIC analysis coordination** — primary analysis liaison between the SKKU group and the Korea-BIC consortium for beam-test data products.
- **KoALICE national activities** — regular participation since 2022, including invited collaboration-meeting talks (see Talks).

Teaching and Mentoring

Student mentoring (non-supervisory) — Sungkyunkwan University 2024 – present

- *Wooseok Ham* (M.Sc., co-main analyzer on the ALICE Run 3 O–O charged-jet measurement) — mentored on Run 3 jet reconstruction methodology, correction procedures, and systematic-uncertainty estimation; co-author with him on the corresponding PWG-JE Analysis Note.
- *[second M.Sc. co-mentee — name TBD]* (co-main analyzer on the same O–O analysis) — co-mentored.
- *Harim Seong* (undergraduate) — ePIC BIC detector-performance studies using JANA2-based reconstruction; energy response and sampling-fraction extraction.

KoALICE O^2 Tutorial — student co-organizer and instructor — Sungkyunkwan University, Suwon2024.01.22 – 23

- Co-organized (with Hyungjun Lee) the two-day Run 3 analysis-framework tutorial for the Korean ALICE community — ∼22 participants from SKKU, Inha University, and other Korean institutes; listed as the official Indico contact for the workshop. indico.cern.ch/event/1361235
- Instructor on *Running O^{2}Physics with Hyperloop* (solo) and co-instructor on *Practical PWG-JE tasks* (with H. Lee).

Teaching Assistant, SKKU Physics (2022 – 2024) — *Nuclear Reactions* (graduate), *Quantum Mechanics II*, *General Physics I and Laboratory I*.

Technical Skills

Languages. English (fluent), Korean (native).
Programming. C++, Python, ROOT, LaTeX, Bash, Git.
HEP frameworks. ALICE O^{2}Physics, Geant4, JANA2 / EIC-recon, PYTHIA, FastJet.
Analysis methods. jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.
Detector hardware. PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.

References

Available on request.